Particle 4.8

Har $d\sigma = \frac{1}{4E_1E_2}\frac{1}{|\vec{v}_1-\vec{v}_2|}|\mathcal{M}_{fi}|^2 d\Pi$

$d\Pi = (2\pi)^4\delta^4(P_F-P_I)\prod_{\text{final states}}\frac{d^3p_i}{(2\pi)^3}\frac{1}{2E_{p_i}}$

$$\Rightarrow d\Pi = (2\pi)^4\delta^4(P_I-P_F)\iint\frac{d^3p'}{(2\pi)^3}\frac{1}{2E_{p'}}\frac{d^3k'}{(2\pi)^3}\frac{1}{2E_{k'}}$$

Kig i CoM så $P_I=0$ & $P_F=p'-k'$

$$\Rightarrow d\Pi = \frac{1}{(2\pi)^2}\int\frac{d^3p'}{4E_{p'}p'}\delta(E_I-E_F)$$

— Note at $E_{p'}$ skifter til $\sqrt{k'^2+m^2}$ her, ig $E_{k'}=p'$ hen bruges med det samme

$$= \frac{1}{(2\pi)^2}\int\frac{p'^2dp'd\Omega}{4E_{p'}p'}\delta\left(\sqrt{p^2+m^2}+p-\sqrt{p'^2+m^2}-p'\right)$$

Kan så bruge at $\delta(g(x)) = \sum_i\frac{\delta(x-x_i)}{|g'(x_i)|}$, $g(x_i)=0$

Har $p_0'=p$ & $g'(p') = \sqrt{p^2+m^2}+p-\sqrt{p'^2+m^2}-p'$

$$\Rightarrow \left|\frac{dg}{dp'}\Big|_{p'=p}\right| = \frac{p'}{p'^2+m^2}+1\Big|_{p'=p} = \frac{p'}{E_{p'}}\Big|_{p'=p}+1 = \frac{p'+E_{p'}}{E_{p'}}\Big|_{p'=p} = \frac{p+E_p}{E_p} = \frac{E_I}{E_p}$$

$$\Rightarrow d\Pi = \frac{1}{(2\pi)^2}\int\frac{p'dp'd\Omega}{4E_{p'}}\frac{E_p}{E_I}\delta(p'-p) = \frac{1}{16\pi^2}\int\frac{p}{E_p}\frac{E_p}{E_I}d\Omega$$

$$= \frac{1}{16\pi^2}\int\frac{p}{E_I}d\Omega \simeq \frac{1}{32\pi^2}\int d\Omega \quad (\text{rel limit } E_I\to 2p)$$

Ignorerer $\int$ ig og ligger på $\frac{d\sigma}{d\Omega}$, får da

$$\frac{d\sigma}{d\Omega} = \frac{1}{4E_pE_k}|\mathcal{M}_{fi}|^2\frac{1}{16\pi^2}\frac{p}{E_I}\frac{1}{|\vec{v}_p-\vec{v}_k|} = \frac{1}{64\pi^2E_p}|\mathcal{M}_{fi}|^2\frac{1}{E_I}\frac{1}{|\vec{v}_p-\vec{v}_k|}$$

med

$$|\vec{v}_p-\vec{v}_k| = \left|\frac{p}{E_p}-\frac{k}{E_k}\right| = \left|\frac{p}{E_p}-\frac{(-p)}{p}\right| = \frac{p}{E_p}+1 = \frac{E_I}{E_p}$$

$$\Rightarrow \frac{d\sigma}{d\Omega} = \frac{1}{64\pi^2}|\mathcal{M}_{fi}|^2\frac{1}{E_pE_I}\frac{E_p}{E_I} = \frac{1}{64\pi^2}|\mathcal{M}_{fi}|^2\frac{1}{E_I^2}$$